

MAJOR PROJECT REGISTRATION FORM

Date:

Project Category:	☐ Research oriented	☐ Application oriented	Batch ID:	CSM-A11
	☐ Product oriented	☐ Other (please specify)		
Specify if others:				

Title of the Project: AI-Enhanced Real-Time Collaborative Platform

Abstract

Real-time collaboration has become an essential requirement in modern distributed, remote, and educational environments, enabling teams to work efficiently on shared digital content. Traditional collaboration systems often encounter challenges such as delayed synchronization, manual conflict resolution, limited visibility of user activities, and difficulties in maintaining consistency among multiple users. These limitations can reduce productivity, coordination, and the overall efficiency of collaborative work. This project presents the design and development of an **AI-Enhanced Real-Time Collaborative Platform** that enables seamless live multi-user interaction with shared digital resources.

The proposed platform allows multiple users to simultaneously create, read, update, and delete (CRUD) shared documents, spreadsheets, and source code while maintaining instant synchronization between participants. To improve the collaborative experience, the system integrates Artificial Intelligence and Machine Learning techniques that provide intelligent features such as content suggestions, activity summarization, usage analytics, and anomaly detection in collaboration patterns. These capabilities help users understand ongoing activities, improve productivity, and support informed decision-making.

The platform also incorporates user presence awareness, real-time communication, version history management, and secure role-based access control. These features provide users with better visibility of team activities while ensuring that shared resources remain secure and manageable. Modern web technologies and real-time communication protocols are used to support efficient interaction, while **Conflict-free Replicated Data Types (CRDTs)** are employed for reliable synchronization and consistency across multiple users.

By combining real-time collaboration, CRDT-based synchronization, and AI/ML-driven insights, the proposed system aims to provide a scalable, reliable, and user-friendly collaborative environment. The platform can be applied across educational institutions, corporate organizations, and software development environments, helping teams collaborate effectively, reduce coordination difficulties, improve productivity, and make better-informed decisions through intelligent analysis of collaborative activities.

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